A

MINI PROJECT REPORT

ON

“TrackMyCash”

SUBMITTED
TO

SAVITRIBAI PHULE PUNE UNIVERSITY

IN PARTIAL FULFILLMENT OF
MASTERS OF COMPUTER APPLICATION
(MCA)

SUBMITTED BY
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UNDER THE GUIDENCE OF
Prof. Pravin S. Pawade

Akole Taluka Education Society’s

AGASTI INSTITUTE OF MANAGEMENT, COMPUTER APPLICATION AND RESEARCH (AIMCAR), AKOLE

2025-2026

DECLARATION

I, hereby declare that, the **Mini Project Report** entitled **"_TiffinServiceFinder"** submitted for the award of the Degree of **Master In Computer Application** under the faculty **Commerce and Management** of **Savitribai Phule Pune University, Pune** is an outcome of my own efforts and a genuine research work done under the guidance of **Prof. Pravin S. Pawade**. I also declare that, this Mini Project Report or any part therein has not been previously submitted by me for the award of any Degree, Diploma, Associateship, Fellowship or Title this or any other University or any other Institution of Higher Learning.

I, further declare that the material obtained from other sources has been duly acknowledged.

Place : Akole
Date : __ / __ / 2025 **Name of student**

ACKNOWLEDGEMENT

This Mini project report is a fruitful outcome of experiences, challenges, and relationships involving many great people to whom I am more indebted than I can possibly acknowledge. Some of them, however, deserve special thanks. I take this opportunity with much pleasure to thank all, who directly and indirectly contributed, helped me for this Mini Project.

I, take this opportunity to express my deep sense of gratitude and indebtedness to my research guide Prof. Pravin S. Pawade *for his keen interest, dynamic guidance and constant encouragement.*

I, express deep sense of gratitude to ***ATES's Agasti Institute of Management, Computer Application and Resrarch (AIMCAR), Akole*** *for his continuous encouragement and support to complete this Mini Project Report.*

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Place : Akole

Date : __ / __ /2025 **Name of student**

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1. Introduction

In today's fast-paced lifestyle, many individuals such as students, small business owners, and working professionals manage multiple financial transactions in their daily lives. Keeping track of money given to or received from different people can become difficult when records are maintained manually in notebooks or remembered mentally. This often leads to confusion, forgotten transactions, and inaccurate financial tracking.

The TrackMyCash application is developed to solve this problem by providing a user-friendly system where users can add customers, record transactions such as "You Gave" and "You Got", and maintain a clear history of all financial activities. The system stores transaction details with date and time, allowing users to track their financial interactions in a structured and convenient manner**.**

1.1 Company Profile / Institute Profile / Client Profile

Institute Profile

This project is developed as a part of the **Master of Computer Applications (MCA) – First Year** academic curriculum. The project is carried out under the guidance of faculty members to provide students with practical knowledge of **web application development, database management, and real-world problem solving**.

The institute focuses on delivering quality technical education by combining theoretical concepts with practical project development. Through this project, students gain hands-on experience in **Java programming, JSP/Servlet development, database connectivity, and web application design**.

Client Profile (Academic Purpose)

The client for this project is considered to be the **academic institute itself**. The system is developed purely for **educational and demonstration purposes** to show how a web-based application can solve real-life problems such as managing daily financial transactions.

The project is not developed for commercial use but serves as a **learning model** that demonstrates how digital ledger systems can replace traditional manual record keeping.

1.2 Abstract

TrackMyCash is a web-based application developed to help users manage and track their daily financial transactions efficiently. In many situations, people lend or borrow money from friends, family members, or customers, and keeping track of these transactions manually in notebooks can lead to confusion or loss of records.

The TrackMyCash system provides a simple digital platform where users can add customers and record financial transactions under two categories: **"You Gave" and "You Got."** The system stores transaction details such as amount, notes, and date, allowing users to maintain a clear history of all financial activities.

The application is developed using **Java (JSP & Servlet), MySQL database, and web technologies such as HTML, CSS, and JavaScript**. The backend manages data storage and retrieval, while the frontend provides a user-friendly interface for easy navigation.

The TrackMyCash system demonstrates how web applications can help individuals and small businesses maintain accurate financial records. The project also focuses on implementing **CRUD operations, relational database design, user authentication, and responsive interface design.**

1.3.1 Existing System

In the existing system, many people manage their daily financial transactions manually using notebooks, paper records, or by remembering transactions mentally. This traditional method is commonly used by small shop owners, students, and individuals who frequently lend or borrow money.

However, manual record keeping is not always reliable. Transactions may be forgotten, records may be misplaced, and calculations may contain errors. Additionally, it becomes difficult to track the total amount given or received from multiple people over time.

Limitations of the Existing System

- High chance of human error in calculations
- Difficulty in tracking multiple transactions
- Records may be lost or damaged
- No centralized digital system for managing transactions

1.3.2 Need for the Proposed System

The proposed TrackMyCash system is designed to overcome the limitations of the manual ledger system. With increasing financial interactions among individuals and small businesses, there is a need for a simple and reliable digital platform to manage transactions.

The TrackMyCash application provides a structured system where users can easily record, update, and view financial transactions. It helps maintain accurate records and reduces the chances of errors.

Advantages of the Proposed System

- Centralized digital platform for managing transactions
- Easy tracking of money given and received
- Quick access to transaction history
- User-friendly and simple interface
- Saves time and reduces manual errors

1.4 Scope of the System

The scope of the **TrackMyCash System** defines the functionalities and boundaries of the project. The system focuses on providing a simple digital platform for managing daily financial transactions between users and their customers.

Scope of the Proposed System

The system allows users to **register and login** securely.
• Users can **add and manage customer details**.
• Users can record transactions under **"You Gave" and "You Got"** categories.
• The system stores transaction details such as **amount, note, date, and time**.
• Users can **view transaction history** for each customer.
• The application provides a **simple and user-friendly interface** for easy navigation.
• The system helps users **track money given and received** from multiple people.
• The project is developed for **educational and academic purposes**.

Limitations of the Scope

The system does not support **online payment integration**.
• Real-time financial analytics are not included.
• Multi-user collaborative accounting features are not implemented.

1.5 Operating Environment – Hardware and Software

The operating environment defines the hardware and software requirements required to develop and run the **TrackMyCash System** efficiently.

1.5.1 Hardware Requirements

The system does not require high-end hardware and can run on a standard personal computer.

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 8 GB
- **Hard Disk:** Minimum 20 GB free space
- **Input Devices:** Keyboard, Mouse
- **Output Devices:** Monitor

1.5.2 Software Requirements

The following software is required for the development and execution of the system:

• Operating System: Windows 10 / Linux
• Programming Language: Java
• Technology: JSP and Servlet
• Database: MySQL
• Web Server: Apache Tomcat
• Frontend Technologies: HTML, CSS, JavaScript
• Development Tool: Eclipse IDE
• Browser: Google Chrome / Mozilla Firefox

1.6 Brief Description of Technology Used

The **TrackMyCash System** is developed using modern web technologies to provide a simple and efficient platform for managing financial transactions. The application is built using **Java**

1.6.1 Operating System Used

The system is developed and executed on the **Windows Operating System**, which provides a stable environment for web application development and database management..

- **Operating System:** Windows 10
- **Alternative Support:** Linux (Unix-based systems)

The project can run on both Windows and Unix-based systems without major changes.

1.6.2 RDBMS / NoSQL Used to Build Database

- The **MySQL Relational Database Management System (RDBMS)** is used to store and manage application data. The database manages structured data related to **users, customers, and transactions**.
- **Database:** MySQL
- **Reason for Choosing PostgreSQL:**
    - Open-source and widely used database system
    - Easy integration with Java applications using JDBC
    - Supports relational data structures
    - Provides high performance and reliability for web applications.

2. Proposed System

The proposed system is TrackMyCash, a web-based application designed to help users manage and track their daily financial transactions digitally. Many individuals, small shop owners, and students often lend or borrow money from others and maintain records manually in notebooks. This manual method can lead to confusion, calculation errors, and loss of records.

2.1 Study of Similar Systems

Before developing the **TrackMyCash System**, several existing methods and systems used for tracking financial transactions were studied.

1. Existing Systems

Mobile Note Applications

Some users store financial records using note-taking apps on their mobile phones.

- Data is not structured
- Difficult to analyze transaction history
- No automatic calculation features

3. Comparison to Proposed System

Feature	Existing Systems	**Proposed System**
Search Method	Manual search, social media, word-of-mouth	Location-based and category-based search
User Interface	Not interactive	Web-based interface (Streamlit)
Real-time Input	Limited	User can input data directly
Deployment	Usually research only	Deployed as a web application
Accuracy	80–87%	Optimized using GridSearchCV and feature selection

4. Conclusion

The review of similar systems shows that traditional methods of tracking financial transactions are mostly manual and unorganized. Users often face difficulties in managing multiple records and maintaining accurate calculations.

The TrackMyCash System solves these problems by providing a structured digital platform where users can easily record, manage, and track their financial transactions.

2.2 Feasibility Study

A feasibility study is conducted to determine whether the TrackMyCash system can be successfully developed and implemented using available resources.

2.2.1 Technical Feasibility

The system is developed using **Java (JSP and Servlet)** technology.
• **MySQL** is used as the database for storing user, customer, and transaction data.
• Frontend technologies such as **HTML, CSS, and JavaScript** are used to create the user interface.
• The system runs on the **Apache Tomcat server**.
• The application can be developed using **Eclipse IDE**.

Conclusion:
The project is technically feasible because all required technologies are widely available and easy to implement.

2.2.2 Operational Feasibility

• Users can easily record and track financial transactions.
• The system provides a simple and user-friendly interface.
• The application reduces manual effort and calculation errors.
• The system can be accessed through a web browser.

2.2.3 Economic / Financial Feasibility

- • The project uses **open-source technologies** such as Java, MySQL, HTML, CSS, and JavaScript.
  • No additional licensing cost is required.
  • Hardware requirements are minimal.

2.2.4 Overall Feasibility

Considering technical, operational, and economic aspects, the **TrackMyCash System** is highly feasible as an academic project. The system provides a practical solution for managing financial records and serves as a useful learning platform for web application development.

2.3 Objectives of Proposed System

The main objective of the **TrackMyCash System** is to develop a simple and reliable platform for managing financial transactions digitally

1. Easy Transaction Management

To allow users to easily record financial transactions such as **money given and money received**.

2. Organized Data Storage

To store customer and transaction data in a structured **database system**.

3. User-Friendly Interface

To provide a simple web interface that allows users to easily navigate and manage records.

4. Accurate Financial Tracking

To reduce errors in manual calculations and provide accurate financial records.

5. Quick Access to Transaction History

To allow users to view past transactions anytime.

6. Scalability

The system can be expanded in the future with additional features such as reports, reminders, and mobile integration.

7. Academic Learning

To help students understand the implementation of **Java-based web applications and database systems**.

2.4 Users of System

The **TrackMyCash System** is designed for different types of users.

1. General Users

• Individuals who want to manage personal financial transactions.
• Can add customers and record money given or received.

2. Small Business Owners

• Shopkeepers or small vendors who maintain credit transactions with customers.
• Can track customer balances and transaction history.

3. Administrator

• Responsible for managing system data and maintaining database records.
• Ensures proper functioning of the system.

3 Analysis and Design

3.1 System Requirements

The system requirements define the functionalities and constraints that the TrackMyCash System must satisfy in order to effectively manage and track financial transactions. These requirements ensure that the system operates correctly and provides useful features to the users.

3.1.1 Functional Requirements

Functional requirements describe the main features and operations of the system.

1. User Registration and Login

• The system should allow users to **register and create an account**.
• Users should be able to **log in securely using valid credentials**.
• The system should validate user information before granting access.

2. Customer Management

• The system should allow users to **add new customers**.
• Users should be able to **view, update, and delete customer details**.
• Each customer should have basic information such as **name, contact number, and notes**.

3. Transaction Management

• The system should allow users to record transactions under **"You Gave" and "You Got."**
• Users should be able to **add transaction details such as amount, date, and notes**.
• The system should allow users to **view transaction history for each customer**.

4. Data Management

• The system should allow users to **update or delete transaction records**.
• The system should maintain accurate financial records in the database.

5. Database Operations

• The system should store and retrieve data efficiently using **MySQL database**.
• It should support **Create, Read, Update, and Delete (CRUD) operations**.

6. Admin Management

• The administrator should be able to **manage system data and users**.
• Admin can monitor the database and ensure proper functioning of the system.

7. User Interface

• The system should provide a **simple and easy-to-use web interface**.
• Users should be able to navigate the system without technical knowledge.

3.1.2 Non-Functional Requirements

Non-functional requirements describe the **quality attributes and performance characteristics** of the system.

1. Performance

• The system should process user requests quickly.
• Transaction data should be retrieved with **minimal response time**.

2. Usability

• The user interface should be **simple, clear, and easy to use**.
• Even users with **basic computer knowledge** should be able to operate the system easily.

3. Reliability

• The system should operate consistently without frequent failures.
• All transaction data should be **stored and retrieved accurately**.

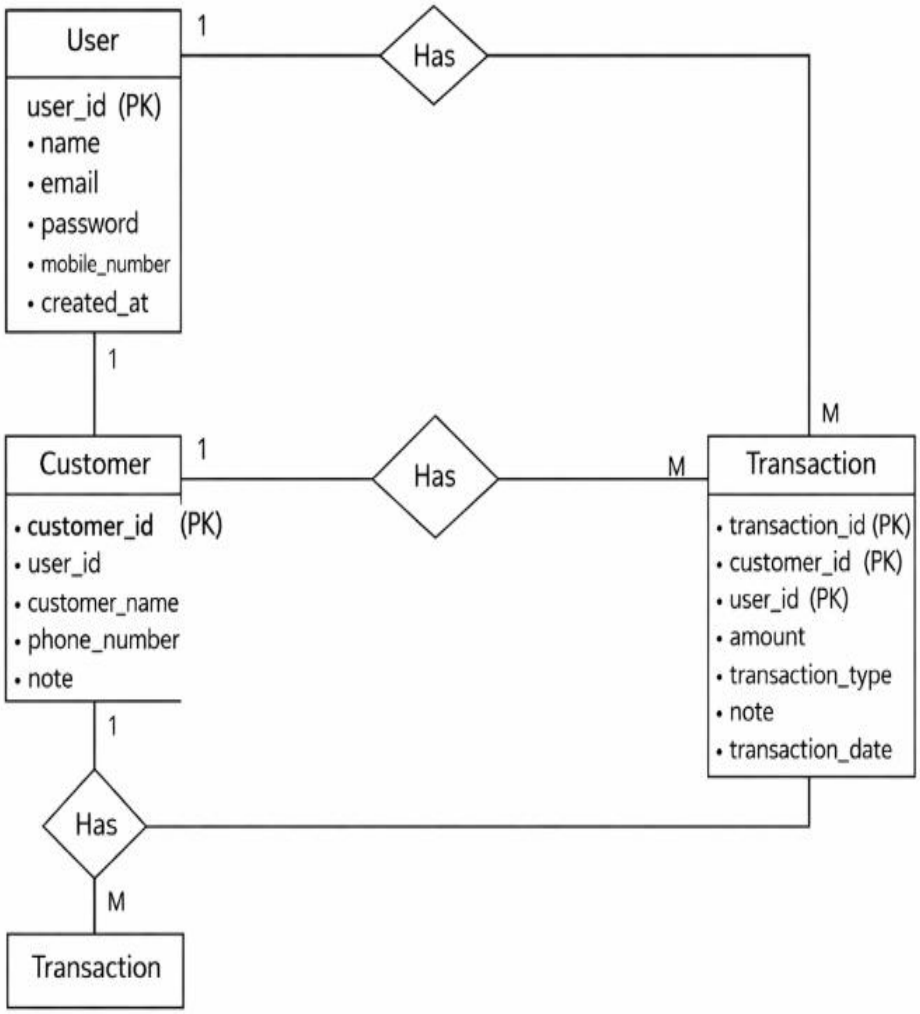
4. Scalability

• The system should be designed to support **future improvements**.
• Additional features such as **financial reports, reminders, and analytics** can be added later.

5. Security

• User login credentials should be **securely stored and validated**.
• Unauthorized users should **not be allowed to access the system**.
• Proper authentication should be implemented to protect user data.

3.2 Entity Relationship Diagram (ERD)



ER Diagr&m for TrackMyCash System

3.3 Table Structure:

User:

Field	Type	Size	Constraints
uid	Integer	11	Primary key
name	Varchar	50	Not null
email	Varchar	255	Not null,Unique
Password	Varchar	255	Not null
mobile_no	Varchar	255	Not null
created_at	TimeStamp	-	Default

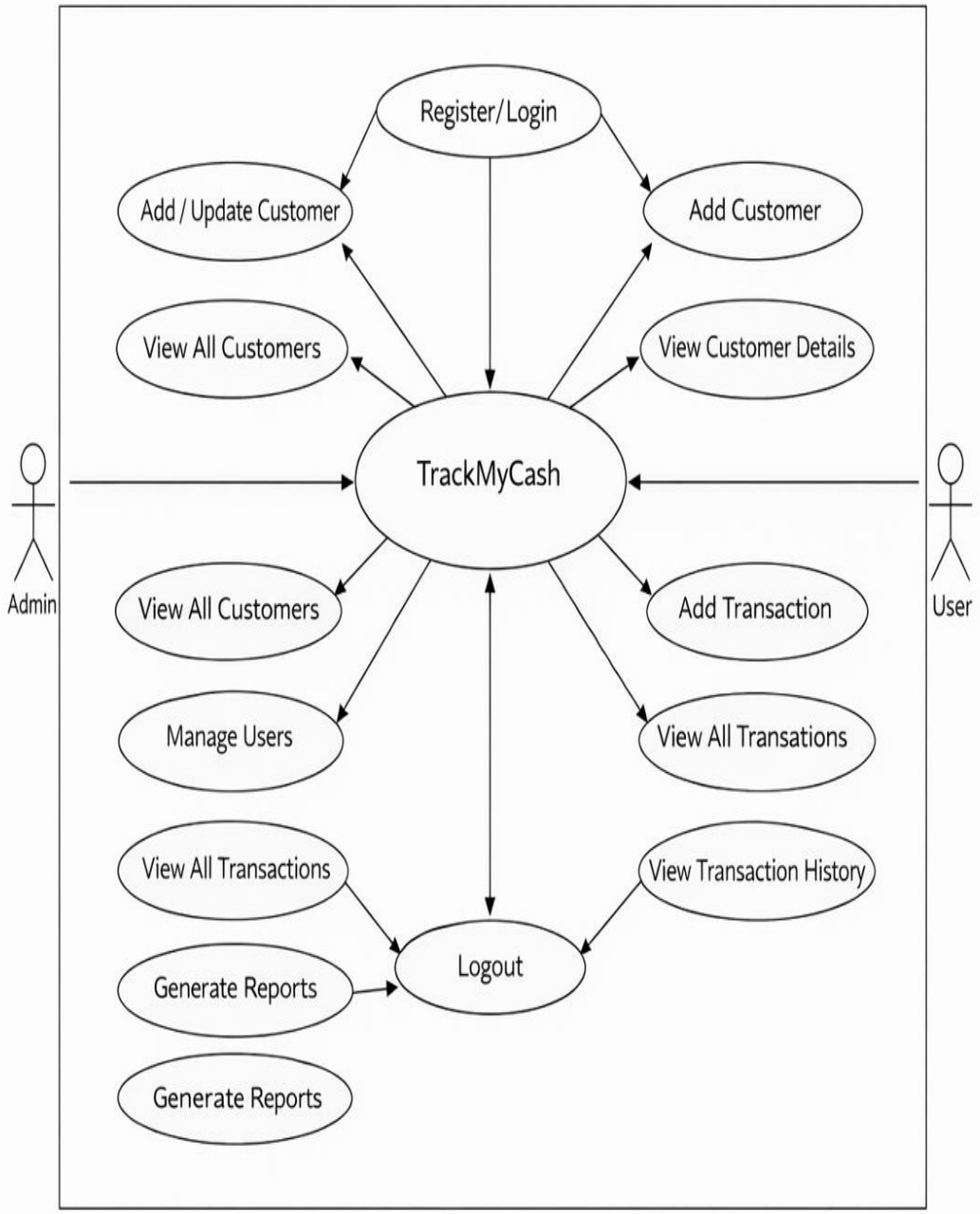
Person:

Field	Type	Size	Constraints
cid	Integer	11	Primary key
uid	Integer	11	Not null
Email	Varchar	100	Not null
Phone_no	Varchar	20	Not null
address	Text	20	Not null
note	Text	10	Not null

Transaction:

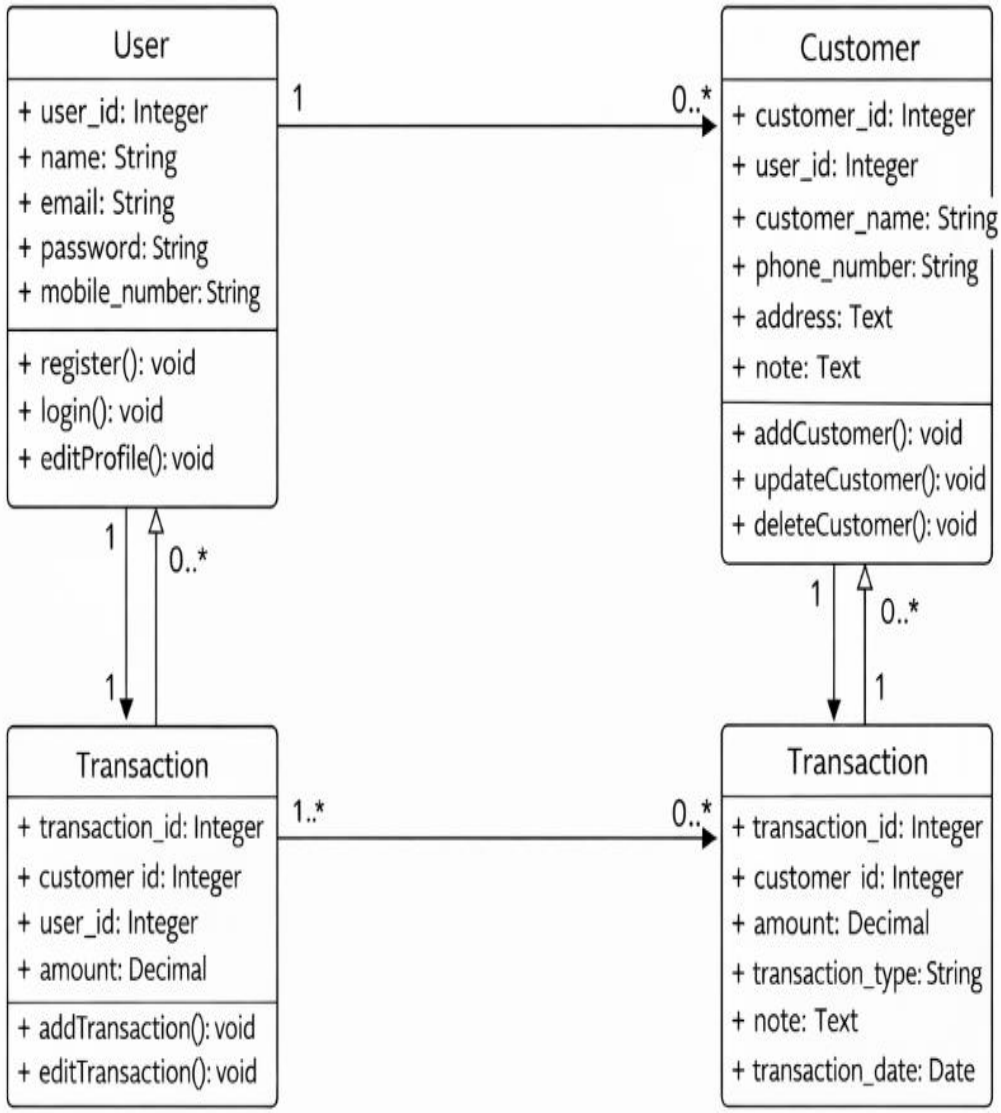
Field	Type	Size	Constraints
tid	Integer	11	Primary key
cid	Integer	11	Foreign key
uid	Integer	11	Not null
amount	Decimal	10,2	Not null
ttype	Varchar	100	Not null
note	Text	-	Not null
tdate	Date	-	Not null

3.4 Use Case Diagrams



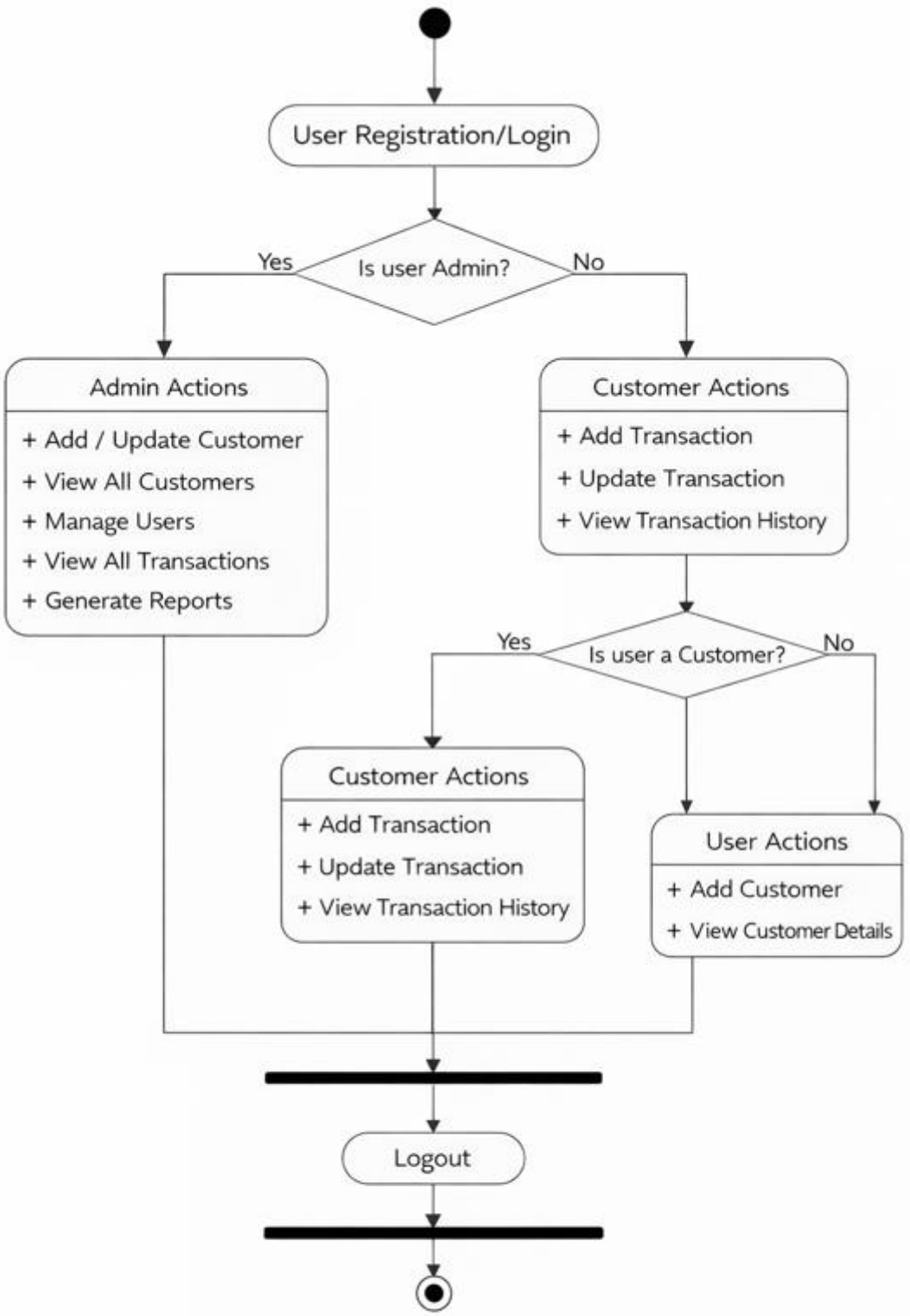
3.4 Use Case Diagrams

3.5 Class Diagram



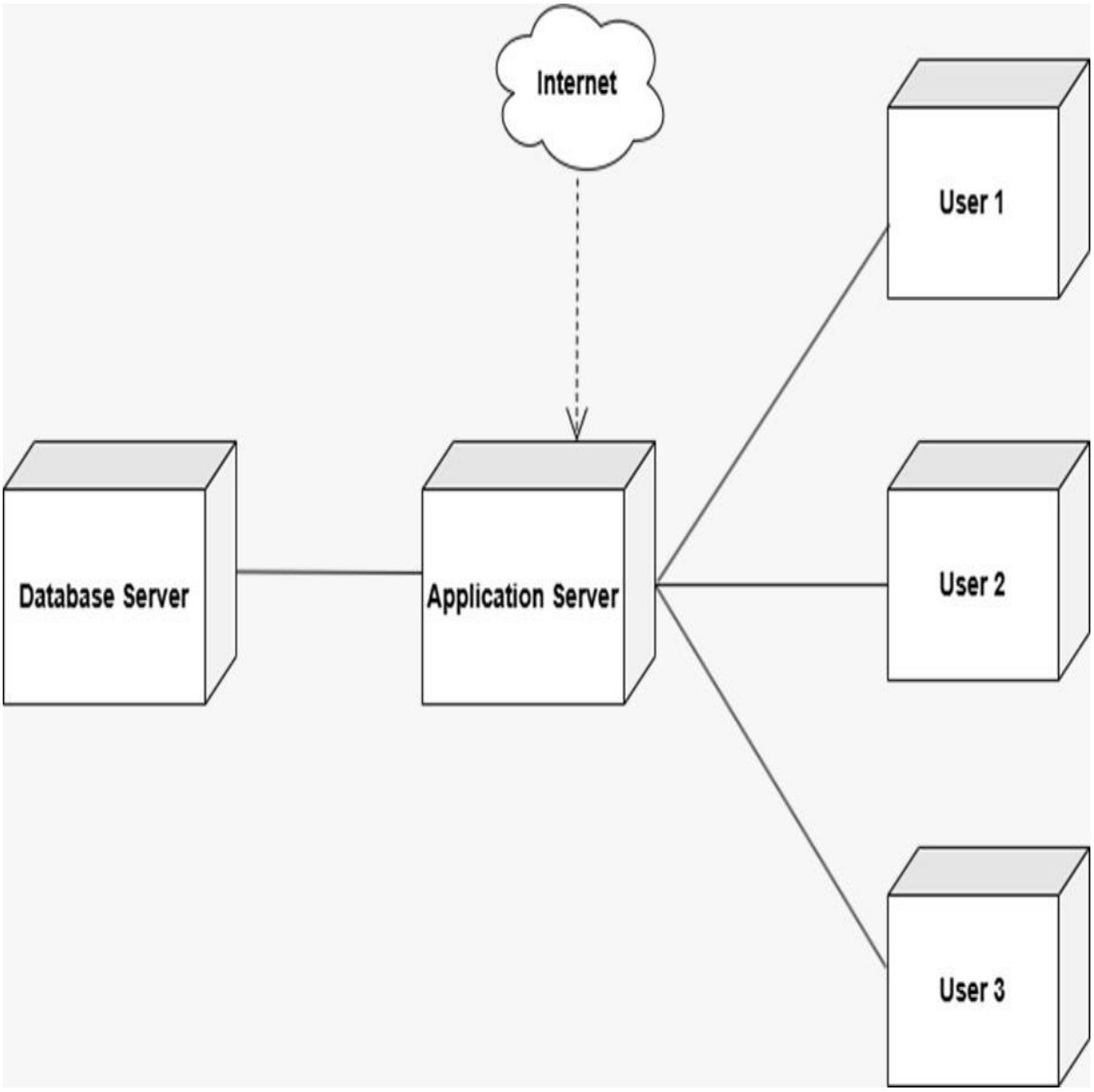
3.4 Class Diagram

3.6 Activity Diagram

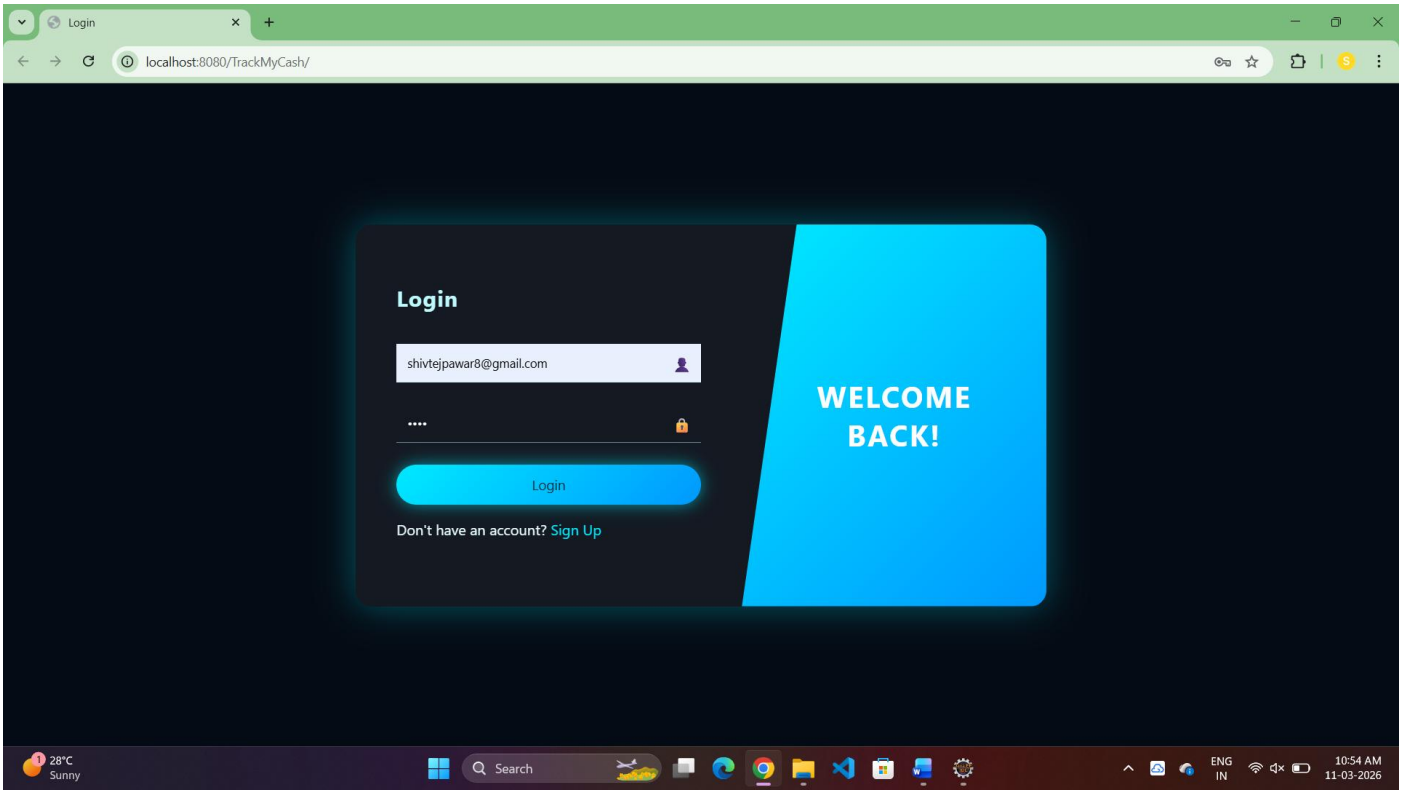


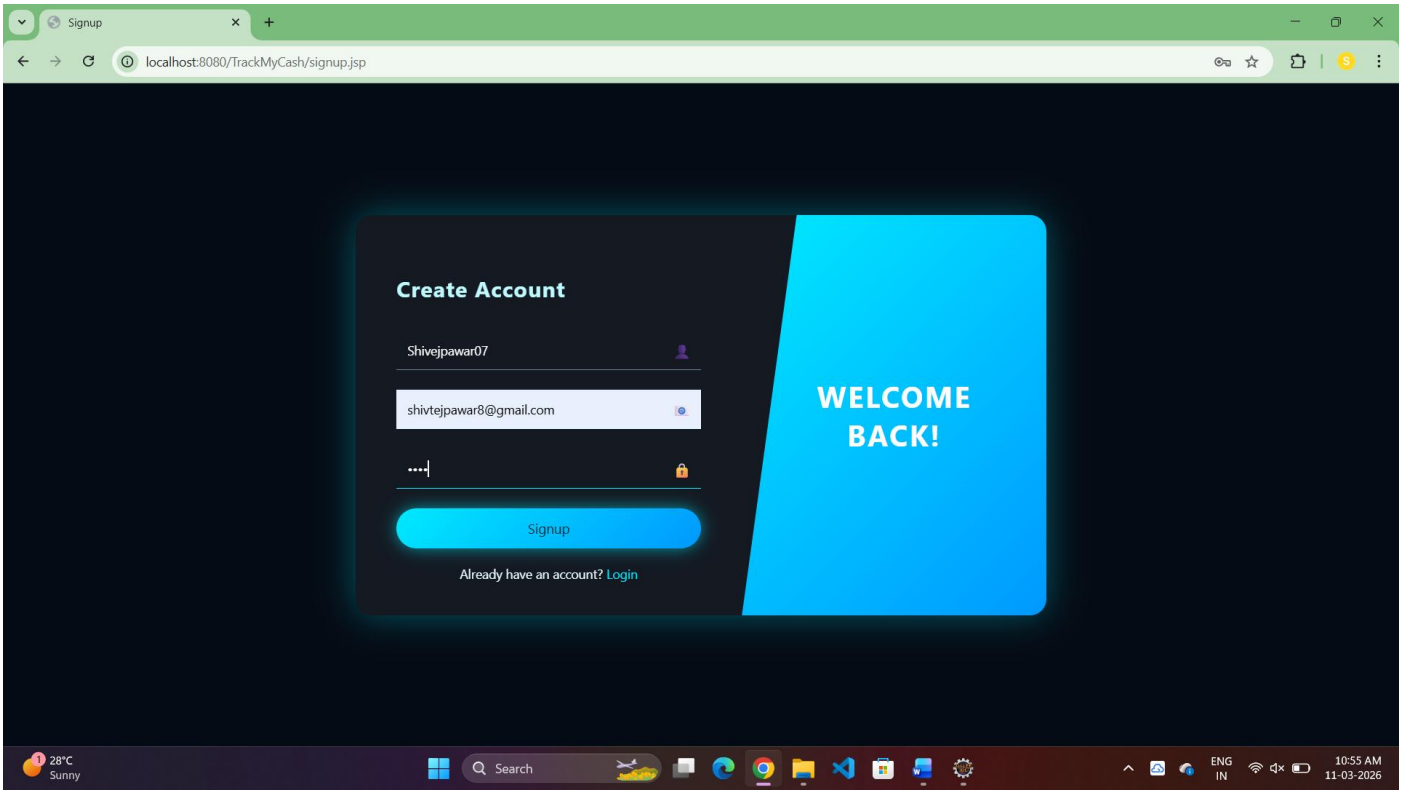
Activity Diagram for TrackMyCash

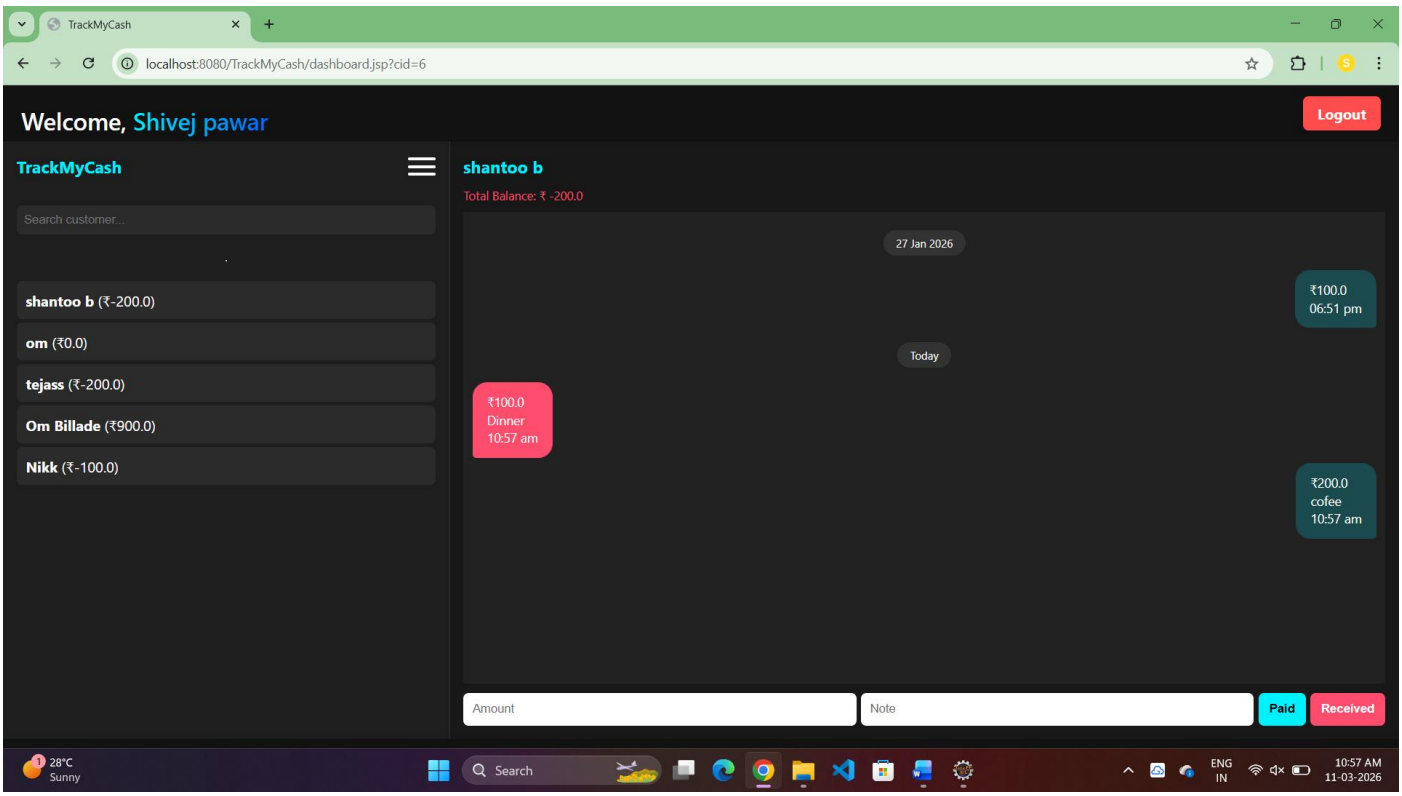
3.7 Deployment Diagram

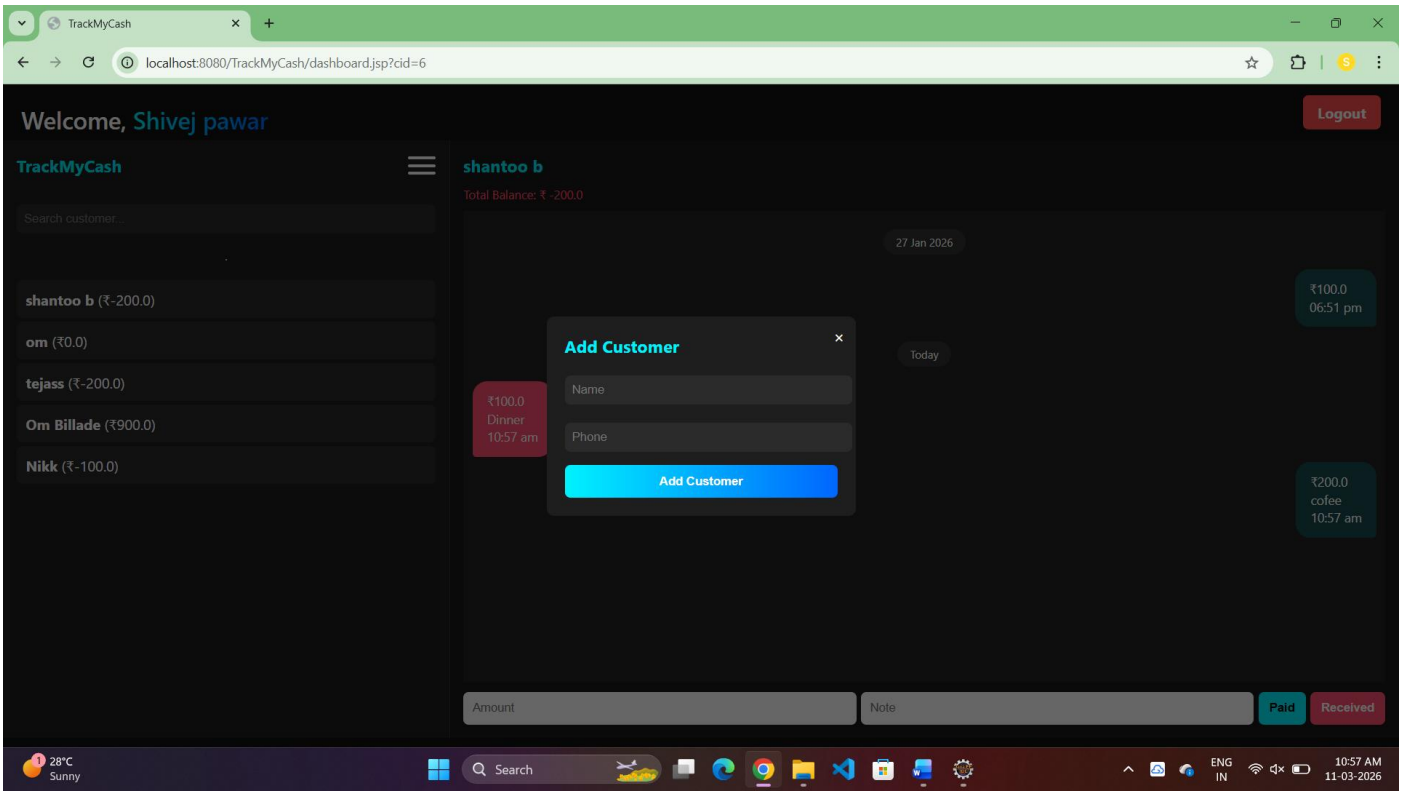


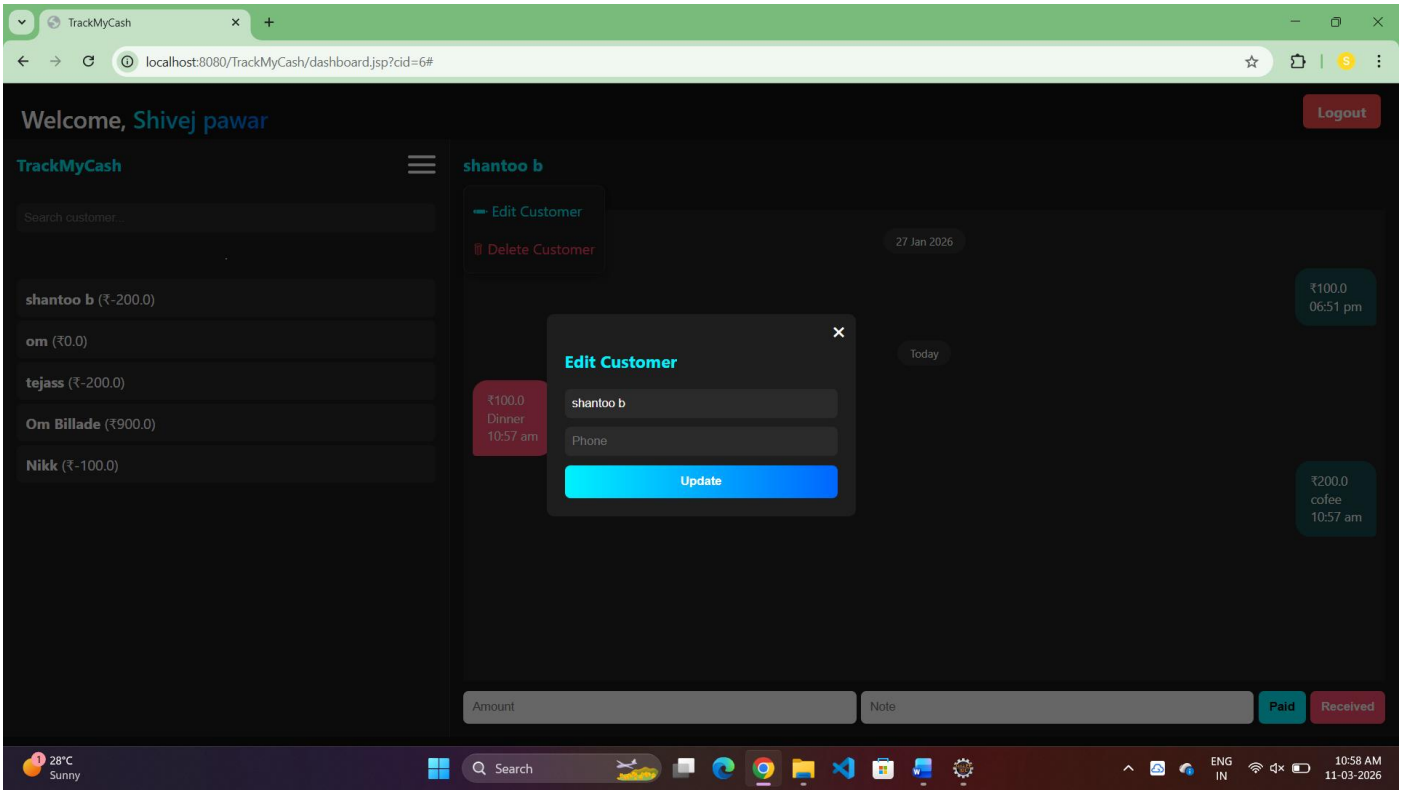
3.9 Sample Input and Output Screens (Screens must have validdata. All reports must have at-least 5 valid records.)




Signup
localhost:8080/TrackMyCash/signup.jsp
Create Account
Shivejpawar07
shivtejpawar8@gmail.com
Signup
Already have an account? Login
WELCOME BACK!
28°C Sunny
Search
ENG IN
10:55 AM 11-03-2026


TrackMyCash
localhost:8080/TrackMyCash/dashboard.jsp?cid=6
Welcome, Shivej pawar
Logout
TrackMyCash
Search customer...
shantoo b (₹-200.0)
om (₹0.0)
tejass (₹-200.0)
Om Billade (₹900.0)
Nikk (₹-100.0)
shantoo b
Total Balance: ₹ -200.0
27 Jan 2026
₹100.0
06:51 pm
Today
₹100.0
Dinner
10:57 am
₹200.0
cofee
10:57 am
Amount
Note
Paid
Received
28°C Sunny
Search
ENG IN
10:57 AM 11-03-2026


TrackMyCash
localhost:8080/TrackMyCash/dashboard.jsp?cid=6
Welcome, Shivej pawar
Logout
TrackMyCash
Search customer...
shantoo b (₹-200.0)
om (₹0.0)
tejass (₹-200.0)
Om Billade (₹900.0)
Nikk (₹-100.0)
shantoo b
Total Balance: ₹ -200.0
27 Jan 2026
₹100.0
06:51 pm
Add Customer
Name
Phone
Add Customer
Today
₹100.0
Dinner
10:57 am
₹200.0
cofee
10:57 am
Amount
Note
Paid
Received
28°C
Sunny
Search
ENG
IN
10:57 AM
11-03-2026


TrackMyCash
localhost:8080/TrackMyCash/dashboard.jsp?cid=6#
Welcome, Shivej pawar
Logout
TrackMyCash
Search customer...
shantoo b (₹-200.0)
om (₹0.0)
tejass (₹-200.0)
Om Billade (₹900.0)
Nikk (₹-100.0)
shantoo b
Edit Customer
Delete Customer
27 Jan 2026
₹100.0
06:51 pm
Edit Customer
shantoo b
Phone
Update
Today
₹100.0
Dinner
10:57 am
₹200.0
cofee
10:57 am
Amount
Note
Paid
Received
28°C
Sunny
Search
ENG
IN
10:58 AM
11-03-2026

4. Coding

This chapter describes the implementation details of the **Track My Cash System**.
The system is developed using **Java, JSP, and Servlet technology** with **MySQL database**.
The application follows a **three-tier architecture**, which separates the presentation layer, business logic layer, and database layer.

4.1 Technologies Used

The following technologies are used in the development of the system:

Technology	Purpose
Java	Backend programming
JSP (Java Server Pages)	Dynamic web pages
Servlet	Handles business logic
MySQL	Database management
HTML, CSS	Frontend design
Bootstrap	Responsive user interface
JavaScript	Client-side validation
Apache Tomcat	Application server

4.2 MVC Architecture Implementation

The system is implemented using **three-layer architecture**.
Presentation Layer (JSP Pages)
Displays the user interface and interacts with the user.
Business Logic Layer (Servlets)
Processes user requests and performs application logic.
Database Layer (MySQL)
Stores and manages financial transaction data.

4.3 Transaction Module

4.3.1 User Module

Purpose:
Allows users to manage their personal financial transactions.
Functions Implemented
- User registration
- User login
- View dashboard
- Logout

4.3.2 Transaction Module

Purpose:
Allows users to track income and expenses.

Functions Implemented:

- Add new transaction
- Update transaction details
- Delete transaction
- View transaction history

4.3.3 Admin Module

Purpose:

Provides administrative control over the system.Functions Implemented:

- Admin login
- Manage users
- View all transactions
- Generate reports

4.4 Database Connectivity Code (MySQL)

```
import java.sql.Connection;
import java.sql.DriverManager;
public class DBConnection {

    public static Connection getConnection() {

        Connection con = null;

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/trackmycash",
                    "root",
                    "7757");

        } catch (Exception e) {
            e.printStackTrace();
        }

        return con;
    }
    }
```

4.5 Servlet Implementation Example

```
String amount = request.getParameter("amount");
String type = request.getParameter("type");
String note = request.getParameter("note");

try {

Connection con = DBConnection.getConnection();
String query = "insert into transactions(amount,type,note) values(?,?,?)";
PreparedStatement ps = con.prepareStatement(query);
ps.setString(1, amount);
ps.setString(2, type);
ps.setString(3, note);
```

```
ps.executeUpdate();
response.sendRedirect("dashboard.jsp");
} catch (Exception e) {
e.printStackTrace();
}
```

4.6 JSP View Implementation Example

```
<h2>Transaction List</h2>
<table border="1">
<tr>
<th>Amount</th>
<th>Type</th>
<th>Note</th>
</tr>
<tr>
<td>500</td>
<td>Expense</td>
<td>Food</td>
</tr>
<tr>
<td>2000</td>
<td>Income</td>
<td>Salary</td>
</tr>
</table>
```

4.8 Sample Output

The system provides the following output:

- Displays list of income and expense transactions
- Shows amount, transaction type, and description
- Allows users to add, update, and delete transactions

4.9 Conclusion

The **Track My Cash System** successfully integrates **JSP, Servlet, and MySQL** technologies to manage personal financial records.
The system allows users to efficiently track income and expenses, maintain transaction history, and manage financial data easily.

The modular structure ensures **easy maintenance and scalability** for future enhancements such as graphical reports and budgeting features.

5. Testing

Testing is an essential phase of software development that ensures the system works correctly and meets the specified requirements.
The Track My Cash System was thoroughly tested using different testing techniques to verify functionality, accuracy, usability, and reliability.

5.1 Test Strategy

The test strategy defines the overall approach used to test the Track My Cash System.
The objective is to ensure that users can easily record income and expenses, view transaction history, and manage financial data without errors.

5.1.1 Testing Approach

The following testing approaches were adopted:

- Manual Testing for user interface and form validation
- Black Box Testing to verify system functionality
- White Box Testing to test internal logic and database queries
- Integration Testing to verify communication between JSP, Servlet, and MySQL

5.1.2 Testing Levels

Testing was carried out at multiple levels:

1. Unit Testing

Individual modules like user registration, login, add transaction, and view transactions were tested separately.

2. Integration Testing

Interaction between JSP pages, Servlets, and MySQL database was verified.

3. System Testing

Complete end-to-end testing of the system including login, transaction management, and logout.

4. Validation Testing

Ensured that the system meets user requirements and functions correctly.

5.1.3 Test Environment

Component	Specification
Operating System	Windows 10
Programming Language	Java
Frontend	HTML, CSS, Bootstrap
Backend	jsp/ Servlet
Database	MySQL
Development Tool	Eclipse

5.1.4 Test Data

- Valid user registration data
- Valid transaction records (Income / Expense)
- Invalid login credentials
- Boundary values (empty fields, large amount values)

5.1.5 Entry and Exit Criteria

Entry Criteria:
- All modules implemented
- Database connected
- Server configured properly

Exit Criteria:
- All test cases executed successfully
- No critical defects remaining

5.1.6 Risk Analysis

Risk	Mitigation Strategy
Incorrect service details	Data validation
Invalid user input	Input validation
Performance delay	Optimized queries
Unauthorized access	Authentication checks

5.1.7 Roles and Responsibilities

- Developer: Coding and unit testing
- Tester: Test case execution
- End User: Acceptance testing

5.2 Unit Test Plan

The Unit Test Plan defines testing of individual modules of the Track My Cash System.

5.2.1 Objectives

- Verify correctness of individual modules
- Detect errors early
- Ensure reliable service search and display

5.2.2 Units to be Tested

Unit ID	Module Name	Description
UT01	Registration Module	User signup
UT02	Login Module	User authentication
UT03	Add Transaction	Record Income expense
UT04	View Transaction	Display Transaction
UT05	Update/Delete	Modify or remove records
UT06	Database Module	Data storage & retrieval

5.2.3 Unit Testing Approach

- White Box Testing
- Independent module testing
- Automated scripts and manual checks

5.2.4 Unit Test Cases

Test Case ID	Module	Input	Expected Output	Status
UT01	Registration	Valid data	Registered	Pass
UT02	Login	Correct credentials	Login success	Pass
UT03	Add	Amount	Saved	Pass
UT04	View	Location	Correct results	Pass
UT05	update	Valid request	Contact shown	Pass
UT06	Database	Insert data	Stored	Pass

5.3 Acceptance Test Plan

Acceptance testing ensures that the system meets user expectations.

5.3.1 Objectives

- Verify ease of use
- Confirm correct service details
- Validate system reliability

5.3.2 Acceptance Criteria

The system is accepted if:

- User can search tiffin services easily
- Correct service details are displayed
- System response time is fast
- No critical defects exist

5.3.3 Acceptance Test Cases

Test Case ID	Scenario	Expected Result	Status
AT01	User Login	Dashboard displayed	Pass
AT02	Add Expense	Expenese Saved	Pass
AT03	View Transaction	Records displayed	Pass
AT04	Delete Transaction	Record Removed	Pass
AT05	LogOut	Session Closed	Pass

5.4 Test Case / Test Script

5.4.1 Sample Test Script

```
public void testAddTransaction() {

    String amount = "500";

   String type = "Expense";

   if(amount != null && type != null) {

      System.out.println("Test Passed");

   } else {

         System.out.println("Test Failed");

   }

}
```

5.4.2 Test Execution Result

- Total Test Cases: 6
- Passed: 6
- Failed: 0

5.5 Defect Report / Test Log

5.5.1 Defect Report Table

Defect ID	Module	Description	Severity	Status
DR01	Login	Invalid Error Msg	Low	Fixed
DR02	UI	Alignment issue	Low	Fixed
DR03	Transaction	Amount Validation	Medium	Fixed
DR04	Database	Duplicate entry	Low	Fixed

5.5.2 Defect Summary

- Total Defects Found: 4
- Critical Defects: 0
- Resolved Defects: 4
- Pending Defects: 0

6. Limitations of Proposed System

Although the Track My Cash System provides an efficient way to manage personal income and expenses, it has certain limitations that should be considered.

6.1 Manual Data Entry

All income and expense records must be entered manually by the user.
If the user forgets to enter transactions, the financial records may become incomplete.

6.2 Internet Dependency

The system is a web-based application, so it requires an active internet connection to access the system and manage transactions.

6.3 Limited Financial Analysis

The system currently provides only basic transaction management.
Advanced financial analysis features such as graphs, reports, and budgeting tools are not included.

6.4 Limited User Access

The system is mainly designed for individual users.
It does not support multi-user financial management such as family or organization expense tracking.

6.5 Basic Security Features

The system includes basic authentication such as login and password validation, but advanced security mechanisms like data encryption, two-factor authentication, and role-based access control are not implemented.

6.6 Scalability Limitations

The current implementation is suitable for small-scale personal usage.
Handling a very large number of users or transactions may require further optimization.

6.7 No Mobile Application

The system is accessible only through a web browser.

A dedicated mobile application is not available in the current version.

7. Proposed Enhancements

The Track My Cash System can be further improved by incorporating additional features and advanced technologies. The following enhancements are proposed to overcome current limitations and improve system performance, usability, and functionality.

7.1 Integration of Advanced Search and Recommendation Techniques

Future versions of the system can use:

Graphical representation of expenses and income
• Monthly and yearly financial reports
• Category-wise expense analysis

TThese features will help users better understand and manage their financial activities.

7.2 Budget Management Feature

The system can be enhanced by allowing tiffin service providers to:

• Set monthly or weekly budgets
• Track spending against the budget
• Receive alerts when expenses exceed the budget limit

This will help users maintain better financial discipline.

7.3 Mobile Application Development

Currently the system is web-based. In the future it can be extended to a mobile application so that users can:

• Record transactions instantly
• Access their financial data anytime and anywhere
• Receive notifications and reminders

7.4 Improved User Interface and User Experience

The system interface can be improved by:

• Interactive dashboards
• Better visualization of transaction history
• Improved navigation and filtering options
• Support for multiple languages such as English, Marathi, and Hindi

7.5 Category-Based Expense Tracking

The system can allow users to categorize transactions such as:

- Food
- Travel
- Shopping
- Bills
- Entertainment

This will make it easier to analyze spending patterns.

7.6 Data Backup and Cloud Integration

Future enhancements may include:

- Automatic data backup
- Cloud storage integration
- Secure data synchronization across devices

7.7 Enhanced Security Features

The system can be strengthened by:

- Password encryption
- Two-factor authentication
- Secure session management
- Role-based access control

7.8 Notification and Reminder System

The system can provide notifications for:

- Upcoming bills or payments
- Monthly expense summaries
- Budget limit warnings

This will help users manage finances more effectively.

7.9 Scalability and Performance Improvements

The system can be expanded to:

- Handle a large number of users
- Optimize database queries
- Improve system performance for large transaction records

8. Conclusion

The Track My Cash System successfully demonstrates the effective use of web-based technology to help users manage their personal financial transactions. The system is designed to allow users to easily record and monitor their income and expenses in a simple and organized manner.

Through proper system analysis, design, implementation, and testing, the application provides a reliable platform for tracking financial activities. Users can add, update, delete, and view their transaction records, which helps them maintain better control over their daily spending.

The system is implemented using JSP, Servlet, and MySQL, which ensures efficient data processing and storage. The user interface is designed to be simple and user-friendly so that users can easily navigate the system and manage their financial data without difficulty.

Various testing methods such as unit testing, integration testing, system testing, and acceptance testing were conducted to ensure the correctness and stability of the application. All identified issues were resolved, ensuring smooth system performance.

Although the system has certain limitations, it provides a useful solution for individuals who want to keep track of their financial activities. With future enhancements such as budget management, graphical reports, mobile application support, and enhanced security features, the system can become a more powerful financial management tool.

In conclusion, the Track My Cash System successfully achieves its objectives and demonstrates how technology can help individuals manage their finances more effectively. The project also provides valuable learning experience in web development using JSP, Servlet, and database integration.

9. Bibliography

- https://codeigniter.com

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  - Beginning PHP 5 , Wrox publication
  - PHP web sevices, Wrox publication
  - AJAX Black Book, Kogent solution
  - Mastering PHP , BPB Publication
  - PHP cookbook, O'Reilly publication

10. Publication / Competition Certificates

This section includes details of publications and competitions related to the Tiffin Service Finder System project. These certificates serve as proof of academic contribution, participation, and recognition.

10.1 Publications

At present, the Tiffin Service Finder System has not been published in any journal or conference. This project has been developed primarily for academic and learning purposes.

(Optional – if published in future):
The project work may be submitted/published in the following journal or conference:

- Title of Paper: Tiffin Service Finder System Using Web Technologies
- Journal / Conference Name: International Journal of Advanced Research in Computer Science (IJARCS) *(Placeholder)*
- Year of Publication: 2025
- Authors: *Shivtej Dipak Pawar*

10.2 Competitions

The Tiffin Service Finder System project has not been submitted to any technical competition so far.

(Optional – if participated in future):
The project may be presented or participated in the following competition:

- Competition Name: Smart City & Local Services Innovation Challenge *(Placeholder)*
- Organized By: XYZ Technical Institute / Tech Foundation *(Placeholder)*
- Year: 2025
- Achievement: Participation

11. Appendix – Cost Sheet, Data Sheet

This appendix provides supplementary information related to the Tiffin Service Finder System, including the estimated cost involved in system development and details of the data used for the project.

11.1 Cost Sheet

The Tiffin Service Finder System is developed using open-source tools and technologies, which significantly reduces the overall development cost. The project is suitable for academic use and small-scale deployment.

Cost Estimation Table

Sr. No.	Resource / Component	Description	Cost (INR)
1	Hardware	Personal Computer / Laptop	Existing
2	Operating System	Windows 10 / Linux	Free
3	Programming Language	PHP / JavaScript / HTML / CSS	Free
4	Framework	CodeIgniter 4	Free
5	Database	MySQL	Free
6	Development Tools	VS Code / XAMPP	Free
7	Internet	Development, testing & research	Minimal
	Total Cost		Low / Nil

Conclusion (Cost Sheet)

The Tiffin Service Finder System is highly cost-effective as it uses open-source software and existing hardware resources. No additional licensing or infrastructure cost is required, making it ideal for academic projects and startup-level applications.

11.2 Data Sheet

The data used in the Tiffin Service Finder System includes details of tiffin service providers and user requirements. The dataset is created manually and/or collected locally for academic purposes.

Dataset Details

- Dataset Name: Tiffin Service Finder Dataset
- Source: Self-created / Local Survey (Academic Purpose)
- Type: Structured data
- Format: MySQL Database / CSV
- Number of Records: Varies (based on available services)

Data Attributes

Attribute Name	Description
Service_ID	Unique tiffin service identifier
Service_Name	Name of tiffin service
Owner_Name	Service provider name
Contact_Number	Contact details
Location	Area / city of service
Meal_Type	Veg / Non-Veg
Price	Monthly or per-meal cost
Delivery_Availability	Yes / No
Rating	User rating
Availability	Active / Inactive

Conclusion (Data Sheet)

The dataset provides essential information required to search, filter, and display tiffin services effectively. Proper data organization and validation ensure reliable system performance and accurate service discovery for users.

12. User Manual – Tiffin Service Finder System

Overview

The Tiffin Service Finder System is a web-based application designed to help users easily find suitable tiffin services based on location, meal type, price, and availability. The system provides a user-friendly interface for searching, viewing, and managing tiffin service details. It also allows service providers to register and manage their services efficiently.

12.1 Login / Authentication Screen

Purpose:

To secure the system and allow only authorized users (Admin, Service Providers, or Registered Users) to access specific features.

Screen Description:

Fields:

1. Username / Email – Enter registered username or email ID
2. Password – Enter password

Validations:

- Username/email cannot be empty
- Password must be at least 6 characters long
- Invalid credentials display error message:
  "Invalid Username or Password."

12.2 Dashboard / Home Screen

Purpose:

To provide easy navigation to all system functionalities.

Screen Description:

Navigation Options:

1. Search Tiffin Services
2. View Tiffin Service List
3. Register Tiffin Service (Provider)

4. My Profile
5. Logout

Validations:

- Dashboard options are accessible only after successful login
- Role-based access is applied (Admin/User/Provider)

12.3 Search Tiffin Service Screen

Purpose:

To allow users to search for available tiffin services.

Screen Description:

Search Filters:

1. Location / City
2. Meal Type – Veg / Non-Veg
3. Price Range
4. Delivery Availability

Validations:

- Location field cannot be empty
- At least one search parameter must be selected
- Error message displayed if no services are found

12.4 Tiffin Service Details Screen

Purpose:

To display complete details of selected tiffin service.

Screen Description:

Displayed Information:

- Service Name
- Owner Name
- Contact Number
- Address / Location
- Meal Type
- Monthly Price
- Delivery Availability
- Special Services

Validations:

- Details are shown only for active services

- Contact details are visible only to logged-in users

12.5 Tiffin Service Registration Screen (Service Provider)

Purpose:

To allow service providers to register their tiffin service.

Screen Description:

Fields:

1. Service Name
2. Owner Name
3. Phone Number
4. Email ID
5. Address / Location
6. Meal Type
7. Price
8. Delivery Option

Validations:

- All mandatory fields must be filled
- Phone number must be numeric and valid
- Email must be in proper format
- Price must be numeric

12.6 Admin Management Screen

Purpose:

To allow admin to manage users and tiffin services.

Screen Description:

Admin Options:

- View all tiffin services
- Approve / Reject service registration
- Activate / Deactivate services
- Manage users

Validations:

- Admin privileges required
- Confirmation popup for delete/deactivate actions

12.7 Reports / Listings Screen

Purpose:

To display tiffin service listings and reports.

Screen Description:

- List of registered tiffin services
- Filters by location, price, or meal type
- Option to export data (optional)

Validations:

- Filters must be valid
- Only existing records can be exported

12.8 Logout / Exit Screen

Purpose:

To safely log out from the system.

Validations:

- Logout confirmation message:
  "Are you sure you want to logout?"
- Session ends after logout
- Redirects user to login page

Conclusion

The Tiffin Service Finder System is simple, user-friendly, and efficient. It allows users to quickly find suitable tiffin services and helps service providers promote their offerings. The system ensures secure access, proper validation, and smooth navigation for all users.